

Hidden Markov Models (HMM): An Interactive and In-Depth Lesson

1. Unveiling the Hidden: An Intuitive Introduction to HMMs

Welcome to an in-depth exploration of **Hidden Markov Models (HMMs)**! These fascinating statistical tools allow us to peek behind the curtain of observable events and infer the hidden processes that generate them. Think of HMMs as sophisticated detectives, piecing together clues from what they see to understand what's *really* going on.

What's the Big Idea? Observable vs. Hidden

At its core, an HMM deals with two types of sequences:

1. **Observable Sequence:** This is what we can directly see or measure. For example, a sequence of words in a sentence, daily stock prices, or a person's mood.
2. **Hidden Sequence:** This is the underlying process or state that we *cannot* directly observe, but which influences the observable sequence. For instance, the grammatical part-of-speech of each word, the true market trend (bullish, bearish), or the weather conditions.

The magic of HMMs lies in their ability to model the relationship between these two sequences, allowing us to make educated guesses about the hidden states based on the observed data.

The Foundation: Revisiting Markov Chains

Before diving deep into the 'hidden' aspect, let's quickly recap **Markov Chains**. A Markov Chain describes a sequence of events where the probability of the next event depends *only* on the current event, and not on the entire history of events. This is known as the **Markov Property**.

Analogy: Imagine a simple weather model where you can directly observe if it's Sunny, Cloudy, or Rainy each day. If tomorrow's weather only depends on today's weather, that's a Markov Chain. You know the current state, and you can predict the next with certain probabilities.

The Leap to HMMs: What if you *couldn't* see the weather directly? What if you only knew whether people carried umbrellas or wore sunglasses? The weather is now *hidden*, and the umbrella/sunglasses observations are *emissions* from those hidden weather states. This is precisely where HMMs come into play!

Visualizing the Difference:

- **Markov Chain:** All states are visible.
- **Hidden Markov Model:** States are hidden, and we only see observations emitted from these hidden states.

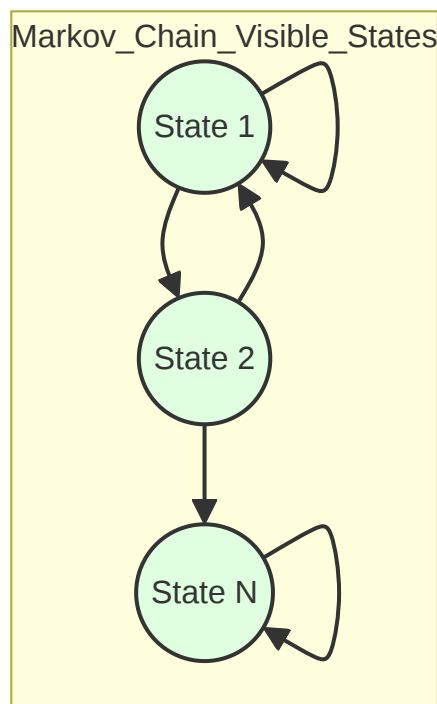


Figure 1: A simple Markov Chain where all states are directly observable.

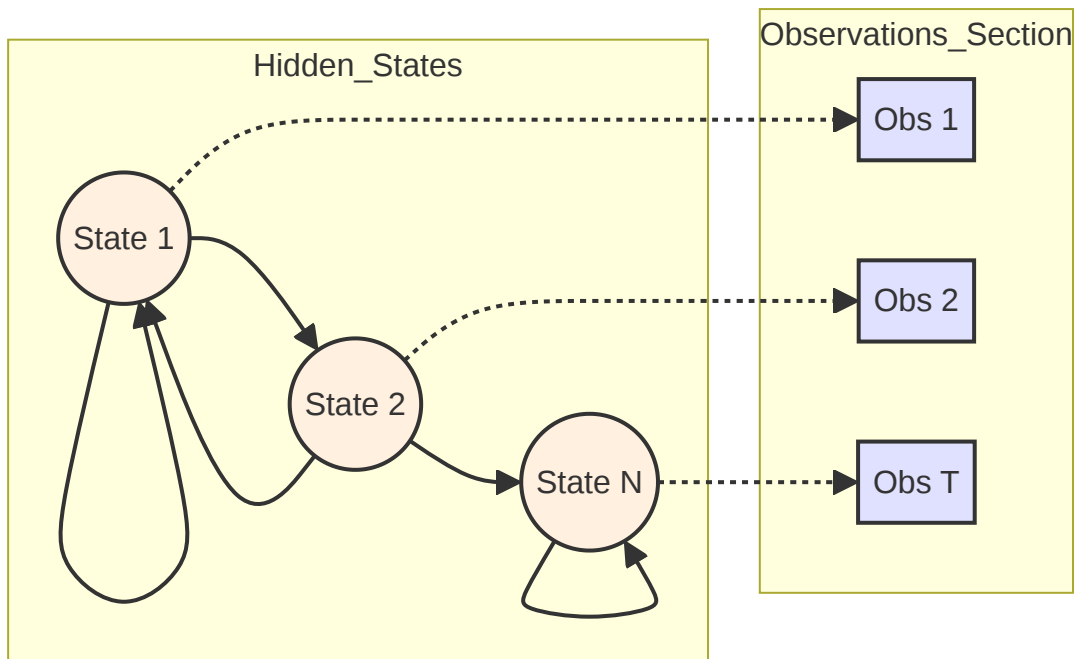


Figure 2: The general architecture of a Hidden Markov Model. Hidden states (S) transition between themselves, and each hidden state emits an observable output (O).

2. The Mathematical Heart of HMMs: Components and Assumptions

To truly understand HMMs, we need to formalize their structure. An HMM is defined by a set of parameters, often collectively denoted as $\lambda = (A, B, \pi)$. Let's break down each component.

2.1. The Five Defining Components of an HMM

1. Number of Hidden States (N):

- This is the total count of distinct hidden states in our model. For example, if our hidden states are 'Hot' and 'Cold' weather, then $N = 2$.
- We denote the set of these states as $S = \{s_1, s_2, \dots, s_N\}$.

2. Number of Observation Symbols (M):

- This is the total count of unique observable events that can be emitted by any hidden state. For instance, if Jason can eat 1, 2, or 3 ice creams, then $M = 3$.

- We denote the set of these observation symbols as $V = \{v_1, v_2, \dots, v_M\}$.

3. State Transition Probability Matrix (A):

- This $N \times N$ matrix, $A = \{a_{ij}\}$, governs how the hidden states evolve over time. Each element a_{ij} represents the probability of transitioning from hidden state s_i to hidden state s_j at the next time step [1].
- **Formula:** $a_{ij} = P(q_{t+1} = s_j | q_t = s_i)$, where $1 \leq i, j \leq N$.
- **Constraint:** The sum of probabilities from any given state must equal 1: $\sum_{j=1}^N a_{ij} = 1$ for all i .
- **Intuition:** If you're in state s_i today, what's the chance you'll be in state s_j tomorrow?

4. Observation (or Emission) Probability Matrix (B):

- This $N \times M$ matrix, $B = \{b_j(k)\}$, describes the likelihood of observing a particular symbol v_k when the system is in a specific hidden state s_j [1].
- **Formula:** $b_j(k) = P(o_t = v_k | q_t = s_j)$, where $1 \leq j \leq N$ and $1 \leq k \leq M$.
- **Constraint:** The sum of probabilities of emitting all possible observations from any single state must equal 1: $\sum_{k=1}^M b_j(k) = 1$ for all j .
- **Intuition:** If the hidden state is s_j , what's the chance you'll observe v_k ?

5. Initial State Probability Distribution (π):

- This is a vector, $\pi = \{\pi_i\}$, where each π_i represents the probability that the HMM starts in hidden state s_i at time $t = 1$ [1].
- **Formula:** $\pi_i = P(q_1 = s_i)$, where $1 \leq i \leq N$.
- **Constraint:** The sum of all initial probabilities must equal 1: $\sum_{i=1}^N \pi_i = 1$.
- **Intuition:** What's the likelihood that the process begins in state s_i ?

2.2. The Foundational Assumptions of HMMs

HMMs are powerful, but they rely on two key simplifying assumptions to make their calculations tractable [2]:

1. Markov Assumption (Limited Horizon):

- The probability of being in a particular hidden state at time t depends *only* on the hidden state at time $t - 1$. It does not depend on any states prior to $t - 1$.
- **Formal:** $P(q_t | q_1, \dots, q_{t-1}) = P(q_t | q_{t-1})$.
- **Why it matters:** This assumption significantly reduces the complexity of modeling the sequence of hidden states.

2. Output Independence Assumption:

- The probability of an observed symbol at time t depends *only* on the hidden state at time t . It is conditionally independent of all other hidden states and all other observations.
- **Formal:** $P(o_t | q_1, \dots, q_T, o_1, \dots, o_T) = P(o_t | q_t)$.
- **Why it matters:** This assumption simplifies the calculation of observation probabilities, as each observation is considered to be generated solely by its corresponding hidden state.

2.3. The Three Fundamental Problems HMMs Solve

With these components and assumptions, HMMs are designed to tackle three core problems [1]:

1. Problem 1: Evaluation (Likelihood):

- **Question:** Given an HMM $\lambda = (A, B, \pi)$ and an observation sequence $O = (o_1, o_2, \dots, o_T)$, what is the probability $P(O | \lambda)$? In other words, how likely is it that this particular HMM generated the observed sequence?
- **Solution Algorithm:** The **Forward Algorithm** (which we will explore in detail).

2. Problem 2: Decoding:

- **Question:** Given an HMM $\lambda = (A, B, \pi)$ and an observation sequence $O = (o_1, o_2, \dots, o_T)$, what is the single most likely sequence of hidden states $Q = (q_1, q_2, \dots, q_T)$ that produced these observations?
- **Solution Algorithm:** The **Viterbi Algorithm**.

3. Problem 3: Learning (Training):

- **Question:** Given an observation sequence $O = (o_1, o_2, \dots, o_T)$ (and sometimes an initial guess for the number of states), how can we adjust the

HMM parameters (A, B, π) to maximize the probability $P(O|\lambda)$?

- **Solution Algorithm:** The **Forward-Backward Algorithm** (also known as the Baum-Welch Algorithm).

In the next section, we will bring these concepts to life with a detailed numerical example, focusing on Problem 1: Evaluation, using the Forward Algorithm.

3. A Deep Dive into the Numerical Example: Jason's Ice Cream and the Weather

Let's make this concrete with a classic and intuitive example: Jason's daily ice cream consumption and the hidden weather. Our goal is to calculate the probability of observing a specific sequence of ice cream counts given a predefined HMM model.

The Scenario: You are an observer. You cannot see the weather (Hot or Cold), but you *can* see how many ice creams Jason eats each day (1, 2, or 3). We want to know: *How likely is a particular sequence of ice cream observations given our understanding of Jason's habits and the weather patterns?*

3.1. Defining Our HMM Parameters (λ)

First, let's set up our HMM $\lambda = (A, B, \pi)$ with specific values:

- **Hidden States (N=2):** $S = \{\text{Hot (H), Cold (C)}\}$
- **Observation Symbols (M=3):** $V = \{1 \text{ ice cream}, 2 \text{ ice creams}, 3 \text{ ice creams}\}$
- **Initial State Probabilities (π):** What's the chance the first day is Hot or Cold?
 - $P(q_1 = \text{Hot}) = 0.8$
 - $P(q_1 = \text{Cold}) = 0.2$
 - So, $\pi = [0.8, 0.2]$. (Notice they sum to 1: $0.8 + 0.2 = 1.0$)
- **State Transition Probabilities (A):** How does the weather change from one day to the next?

From \ To	Hot (H)	Cold ©
Hot (H)	0.6	0.4
Cold ©	0.5	0.5

- $a_{H,H} = P(q_{t+1} = \text{Hot} | q_t = \text{Hot}) = 0.6$
- $a_{H,C} = P(q_{t+1} = \text{Cold} | q_t = \text{Hot}) = 0.4$
- $a_{C,H} = P(q_{t+1} = \text{Hot} | q_t = \text{Cold}) = 0.5$
- $a_{C,C} = P(q_{t+1} = \text{Cold} | q_t = \text{Cold}) = 0.5$
- Notice rows sum to 1: $0.6 + 0.4 = 1.0$, $0.5 + 0.5 = 1.0$.

- **Observation (Emission) Probabilities (B):** How many ice creams does Jason eat given the weather?

State \ Obs	1 Ice Cream	2 Ice Creams	3 Ice Creams
Hot (H)	0.2	0.4	0.4
Cold ©	0.5	0.4	0.1

- $b_H(1) = P(o_t = 1 | q_t = \text{Hot}) = 0.2$
- $b_H(2) = P(o_t = 2 | q_t = \text{Hot}) = 0.4$
- $b_H(3) = P(o_t = 3 | q_t = \text{Hot}) = 0.4$
- $b_C(1) = P(o_t = 1 | q_t = \text{Cold}) = 0.5$
- $b_C(2) = P(o_t = 2 | q_t = \text{Cold}) = 0.4$
- $b_C(3) = P(o_t = 3 | q_t = \text{Cold}) = 0.1$
- Notice rows sum to 1: $0.2 + 0.4 + 0.4 = 1.0$, $0.5 + 0.4 + 0.1 = 1.0$.

Visualizing Our Specific HMM:

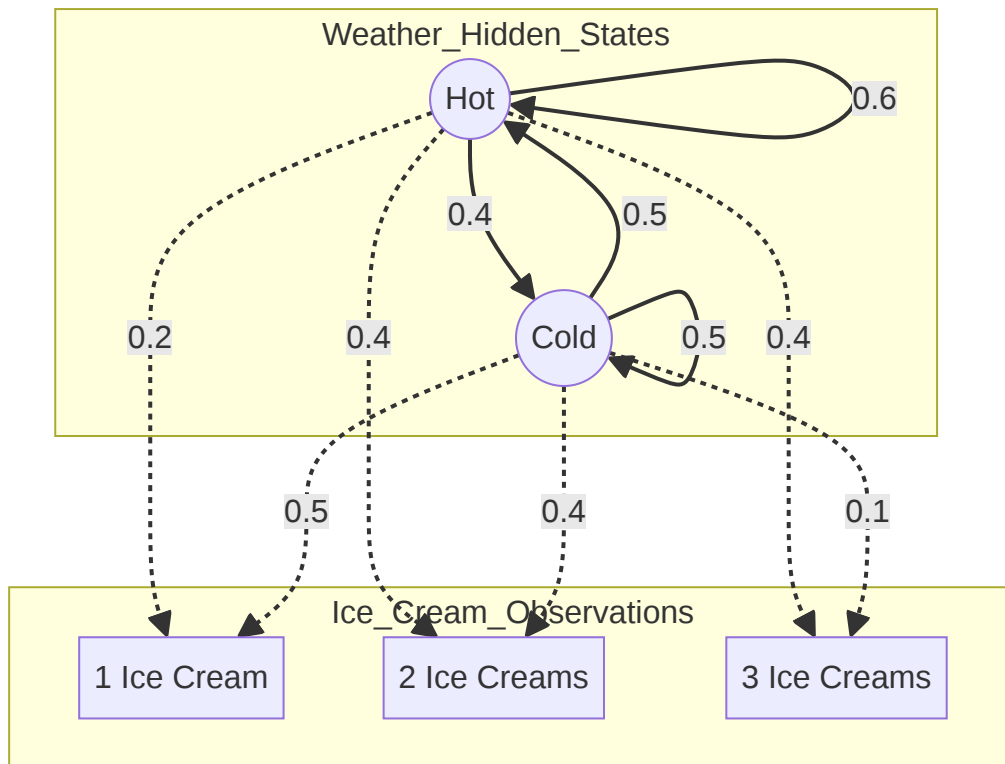


Figure 3: HMM for the Ice Cream and Weather example. Solid arrows represent state transitions, and dashed arrows represent emissions. The numbers on the arrows are the probabilities.

3.2. The Problem: Calculating the Likelihood of an Observation Sequence

Let's say we observe Jason's ice cream consumption for three consecutive days:

Observation Sequence: $O = (o_1 = 3, o_2 = 1, o_3 = 3)$

This means:

- Day 1: Jason ate 3 ice creams.
- Day 2: Jason ate 1 ice cream.
- Day 3: Jason ate 3 ice creams.

Our task is to calculate $P(O|\lambda)$, the probability of observing this specific sequence given our HMM. Why is this hard? Because we don't know the *hidden* weather sequence! It could be (Hot, Hot, Hot), (Hot, Cold, Hot), (Cold, Hot, Cold), and so on. There are N^T possible hidden state sequences, which can be a huge number.

Manually calculating the probability for each possible hidden state sequence and summing them up would be computationally infeasible for longer sequences. This is where the **Forward Algorithm** comes to our rescue.

3.3. The Forward Algorithm: A Dynamic Programming Approach

The Forward Algorithm efficiently calculates $P(O|\lambda)$ using dynamic programming. It works by building up partial probabilities over time. Let's define a key variable:

Forward Variable ($\alpha_t(i)$):

$\alpha_t(i)$ represents the probability of observing the partial sequence $o_1, \dots, o_t$ AND being in hidden state s_i at time t . Formally:

$$\alpha_t(i) = P(o_1, \dots, o_t, q_t = s_i | \lambda)$$

We can break the Forward Algorithm into three steps:

Step 1: Initialization (t=1)

For the first observation o_1 , the probability of being in state s_i and emitting o_1 is simply the initial probability of state s_i multiplied by the emission probability of o_1 from s_i .

Formula: $\alpha_1(i) = \pi_i \times b_i(o_1)$

Let's calculate for our example, with $o_1 = 3$:

- **For Hidden State = Hot (H):** $\alpha_1(\text{Hot}) = \pi_{\text{Hot}} \times b_{\text{Hot}}(o_1 = 3)$ $\alpha_1(\text{Hot}) = 0.8 \times 0.4 = 0.32$ *Interpretation:* There's a 32% chance that the first day was Hot AND Jason ate 3 ice creams.
- **For Hidden State = Cold ©:** $\alpha_1(\text{Cold}) = \pi_{\text{Cold}} \times b_{\text{Cold}}(o_1 = 3)$ $\alpha_1(\text{Cold}) = 0.2 \times 0.1 = 0.02$ *Interpretation:* There's a 2% chance that the first day was Cold AND Jason ate 3 ice creams.

Step 2: Recursion (t=2 to T)

For each subsequent time step t (from 2 to T), and for each possible hidden state s_j at time t , we calculate $\alpha_t(j)$. This involves summing over all possible previous states s_i at time $t - 1$. For each previous state s_i , we consider:

1. The probability of having reached s_i at $t - 1$ with the partial observation sequence ($\alpha_{t-1}(i)$).
2. The probability of transitioning from s_i to s_j (a_{ij}).
3. The probability of emitting the current observation o_t from state s_j ($b_j(o_t)$).

Formula: $\alpha_t(j) = \left[\sum_{i=1}^N \alpha_{t-1}(i) \times a_{ij} \right] \times b_j(o_t)$

Let's apply this for our example:

For t=2 (Observation $o_2 = 1$):

- **For Hidden State = Hot (H) at t=2:** $\alpha_2(\text{Hot}) = [\alpha_1(\text{Hot}) \times a_{\text{Hot,Hot}} + \alpha_1(\text{Cold}) \times a_{\text{Cold,Hot}}] \times b_{\text{Hot}}(o_2 = 1)$
 $\alpha_2(\text{Hot}) = [0.32 \times 0.6 + 0.02 \times 0.5] \times 0.2$
 $\alpha_2(\text{Hot}) = [0.192 + 0.01] \times 0.2$
 $\alpha_2(\text{Hot}) = 0.202 \times 0.2 = 0.0404$ *Interpretation:* There's a 4.04% chance that the sequence (3, 1) was observed AND the weather on Day 2 was Hot.
- **For Hidden State = Cold © at t=2:** $\alpha_2(\text{Cold}) = [\alpha_1(\text{Hot}) \times a_{\text{Hot,Cold}} + \alpha_1(\text{Cold}) \times a_{\text{Cold,Cold}}] \times b_{\text{Cold}}(o_2 = 1)$
 $\alpha_2(\text{Cold}) = [0.32 \times 0.4 + 0.02 \times 0.5] \times 0.5$
 $\alpha_2(\text{Cold}) = [0.128 + 0.01] \times 0.5$
 $\alpha_2(\text{Cold}) = 0.138 \times 0.5 = 0.069$ *Interpretation:* There's a 6.9% chance that the sequence (3, 1) was observed AND the weather on Day 2 was Cold.

For t=3 (Observation $o_3 = 3$):

- **For Hidden State = Hot (H) at t=3:** $\alpha_3(\text{Hot}) = [\alpha_2(\text{Hot}) \times a_{\text{Hot,Hot}} + \alpha_2(\text{Cold}) \times a_{\text{Cold,Hot}}] \times b_{\text{Hot}}(o_3 = 3)$
 $\alpha_3(\text{Hot}) = [0.0404 \times 0.6 + 0.069 \times 0.5] \times 0.4$
 $\alpha_3(\text{Hot}) = [0.02424 + 0.0345] \times 0.4$
 $\alpha_3(\text{Hot}) = 0.05874 \times 0.4 = 0.023496$ *Interpretation:* There's a 2.35% chance that the sequence (3, 1, 3) was observed AND the weather on Day 3 was Hot.
- **For Hidden State = Cold © at t=3:** $\alpha_3(\text{Cold}) = [\alpha_2(\text{Hot}) \times a_{\text{Hot,Cold}} + \alpha_2(\text{Cold}) \times a_{\text{Cold,Cold}}] \times b_{\text{Cold}}(o_3 = 3)$
 $\alpha_3(\text{Cold}) = [0.0404 \times 0.4 + 0.069 \times 0.5] \times 0.1$
 $\alpha_3(\text{Cold}) = [0.01616 + 0.0345] \times 0.1$
 $\alpha_3(\text{Cold}) = 0.05066 \times 0.1 = 0.005066$ *Interpretation:* There's a 0.51% chance that the sequence (3, 1, 3) was observed AND the weather on Day 3 was Cold.

Step 3: Termination (Total Probability)

Once we have calculated $\alpha_T(i)$ for all hidden states s_i at the final time step T , the total probability of the observed sequence O given the model λ is simply the sum of these final forward probabilities.

Formula: $P(O|\lambda) = \sum_{i=1}^N \alpha_T(i)$

Let's complete our example:

$$P(O = (3, 1, 3)|\lambda) = \alpha_3(\text{Hot}) + \alpha_3(\text{Cold}) \qquad P(O = (3, 1, 3)|\lambda) = 0.023496 + 0.005066 = 0.028562$$

So, the probability of observing Jason eat (3, 1, 3) ice creams over three days, given our HMM, is approximately **0.028562** (or about 2.86%).

Visualizing the Forward Algorithm: The Trellis Diagram

The calculations of the Forward Algorithm can be beautifully visualized using a **Trellis Diagram**. This diagram shows all possible paths through the hidden states over time, along with the accumulated probabilities at each step.

 HMM Trellis Diagram (Ice Cream Example)

Figure 4: Trellis diagram illustrating the Forward Algorithm for the Ice Cream example. Each node represents a hidden state at a given time, and the values within or near the nodes are the $\alpha_t(i)$ probabilities. The paths show transitions and emissions contributing to these probabilities.

Self-reflection: How does this diagram help us understand the dynamic programming aspect? Notice how each $\alpha_t(i)$ value is calculated by combining values from the previous time step, avoiding redundant calculations.

4. Why Hidden Markov Models Are Indispensable

HMMs are not just theoretical constructs; they are incredibly powerful tools that have shaped many technological advancements. Their importance stems from several key attributes:

- **Uncovering the Unseen:** HMMs provide a rigorous mathematical framework for inferring hidden causes from observable effects. This ability to model latent processes is crucial in fields where direct observation is impossible or impractical.
- **Robustness in Sequential Data:** They are specifically designed for sequential data, effectively capturing temporal dependencies and patterns that evolve over time. This makes them ideal for time-series analysis.
- **Probabilistic Rigor:** HMMs offer a probabilistic foundation, allowing us to quantify uncertainty, calculate likelihoods, and make informed decisions based on probabilities rather than deterministic rules.
- **Foundation for Advanced AI:** HMMs laid much of the groundwork for modern sequence modeling techniques, including those in deep learning (e.g., Recurrent Neural Networks, LSTMs). Understanding HMMs provides a strong conceptual basis for these more complex models.
- **Interpretability:** Unlike some black-box models, the parameters of an HMM (transition and emission probabilities) often have intuitive interpretations, offering insights into the underlying system dynamics.

5. Real-life Applications: Where HMMs Shine

HMMs have a rich history of successful applications across diverse domains. Here are some prominent examples that highlight their versatility [2] [3] [4]:

1. **Speech Recognition:** This is perhaps the most famous application. When you speak into your phone or smart speaker, HMMs are often at work. The observed data are the acoustic signals (sound waves), and the hidden states are the phonemes or words being spoken. HMMs determine the most probable sequence of words that generated the sounds.
2. **Natural Language Processing (NLP):**
 - **Part-of-Speech (POS) Tagging:** HMMs can assign the correct grammatical tag (e.g., noun, verb, adjective) to each word in a sentence. The words are observations, and the tags are hidden states.
 - **Machine Translation:** Used to model the alignment between words or phrases in different languages.
 - **Information Extraction:** Identifying specific entities (like names of people, organizations, or locations) within unstructured text.

3. **Bioinformatics:** HMMs are indispensable for analyzing biological sequences.
 - **Gene Finding:** Identifying coding regions (genes) within vast DNA sequences. The hidden states represent whether a segment of DNA is part of a gene or not, and the observations are the DNA bases (A, T, C, G).
 - **Protein Structure Prediction:** Predicting the secondary structure of proteins (e.g., alpha-helix, beta-sheet) from their amino acid sequences.
 - **Sequence Alignment:** Aligning DNA or protein sequences to find similarities and evolutionary relationships, crucial for understanding evolution and function.
4. **Financial Modeling:** HMMs can be used to model and predict market behavior. Hidden states might represent different market regimes (e.g., bull market, bear market, stagnant market), and observations could be stock prices, trading volumes, or economic indicators.
5. **Activity Recognition:** In smart homes, wearable devices, or surveillance systems, HMMs can recognize human activities (e.g., walking, running, sitting, sleeping) from sensor data (e.g., accelerometer readings, GPS data). The activities are the hidden states, and the sensor readings are the observations.
6. **Medical Diagnostics:** Modeling the progression of diseases, where hidden states could be different stages of a disease (e.g., early, moderate, severe), and observations are symptoms, test results, or vital signs. This can help in early diagnosis and treatment planning.
7. **Robotics:** For tasks like robot localization and mapping, where the robot's true position (hidden state) is inferred from noisy sensor readings (observations) from cameras, lidar, or odometry.

These diverse applications underscore the power and flexibility of Hidden Markov Models in deciphering complex sequential data where the underlying generative mechanisms are not directly observable.

References

- [1] Jurafsky, D., & Martin, J. H. (2026). *Speech and Language Processing*. Draft of January 6, 2026. <https://web.stanford.edu/jurafsky/slp3/A.pdf> [2] Christopher, V. V. (2024). *Understanding the Hidden Markov Model*. Built In. <https://builtin.com/articles/hidden-markov-model> [3] GeeksforGeeks. (n.d.). *Hidden Markov Model in Machine learning*. <https://www.geeksforgeeks.org/machine-learning/hidden-markov-model-in-machine->

[learning/](#) [4] Ultralytics. (n.d.). *What is a Hidden Markov Model (HMM)? Guide*.
<https://www.ultralytics.com/glossary/hidden-markov-model-hmm>